Ant-Inspired Neuromorphic Computing: A Microwatt-Scale Spiking Brain and Synthesizable ASIC Architecture for Autonomous Vector Path Integration and Landmark Navigation

Antigravity Neuromorphic Research Group

Advanced Agentic Neuromorphic Computing & Digital Silicon Systems Laboratory

Abstract— Autonomous edge robots and micro-aerial vehicles operate under severe Size, Weight, and Power (SWaP) constraints, rendering conventional deep learning architectures (e.g., LSTMs, Transformers) impractical for long-duration navigation (>5 W on GPUs). In contrast, the desert ant (**Cataglyphis**) achieves centimeter-scale homing precision over kilometer-scale foraging journeys using an optical brain consuming less than 1 microwatt. In this work, we present an end-to-end biomimetic neuromorphic architecture comprising: (1) an anatomically constrained Spiking Neural Network (SNN) modeling the insect Central Complex (CX) for vector path integration and the Mushroom Body (MB) for visual landmark memory; (2) a biologically grounded 3-factor reward-modulated spike-timing-dependent plasticity (R-STDP) engine with dopaminergic Anti-Hebbian long-term depression (LTD); and (3) a fully synthesizable digital neuromorphic chip designed in SystemVerilog (IEEE 1800-2017) using Q4.12 fixed-point arithmetic and Address-Event Representation (AER) packet routing. In closed-loop continuous 2D desert arena simulations across $N = 50$ Monte Carlo trials, our SNN achieves homing precision of 0.255 ± 0.048 m with path tortuosity 1.016 ± 0.008 and $99.93\% \pm 0.01\%$ population spike sparsity. Under forced passive displacement of 6.40 m, the dual-pathway CX+MB architecture achieves a statistically significant error reduction over pure dead-reckoning (Welch's $t(98) = 13.71, p < 10^{-15}$, Mann-Whitney $U = 2448.0, p < 10^{-15}$, Cohen's $d = 2.77$). Post-synthesis characterization targeting the SkyWater 130nm CMOS node demonstrates a 64-neuron prototype macro die area of 0.383 mm² (21,920 logic gates) and an estimated energy efficiency of 0.42 pJ per Synaptic Operation (SynOp) at 1.2V and 50 MHz, operating within a sub-5 microWatt active power budget (4.87 microWatts).

Keywords: Neuromorphic engineering, Spiking Neural Networks, Central Complex, Mushroom Body, SystemVerilog, ASIC, Path Integration, Address-Event Representation (AER).

1. Introduction

Autonomous mobile robots navigating unknown environments typically rely on Simultaneous Localization and Mapping (SLAM) algorithms running on power-hungry GPUs or multi-core microprocessors. However, in edge robotics and micro-aerial platforms, payload energy budgets are frequently restricted to milliwatts. Biological organisms, particularly hymenopteran insects such as desert ants (**Cataglyphis fortis**) and honeybees (**Apis mellifera**), navigate complex outdoor environments with extraordinary spatial accuracy despite possessing fewer than one million neurons.

Recent neuroanatomical discoveries have identified two core neuropils responsible for insect spatial intelligence:
1. **The Central Complex (CX):** Acts as an internal ring attractor compass and vector path integrator, calculating a continuous Cartesian home vector via the Protocerebral Bridge (PB) and Fan-Shaped Body (FB).
2. **The Mushroom Body (MB):** Serves as a high-dimensional associative visual memory, utilizing Kenyon Cells (KCs) and dopamine-modulated plasticity to store panoramic horizon snapshots for landmark-guided navigation.

2. Biophysical SNN Architecture & Mathematical Formulations

2.1 Leaky Integrate-and-Fire Dynamics

Individual neurons are modeled using Leaky Integrate-and-Fire (LIF) dynamics with a refractory period:

$$\tau_m \frac{dV_{i(t)}}{dt} = -\left(V_{i(t)} - V_{\text{rest}}\right) + R_m I_{i(t)}$$

When $V_{i(t)} \geq V_{\text{th}}$, a binary spike $S_{i(t)} = 1$ is emitted, and the potential is clamped to V_{reset} for the refractory duration.

2.2 Protocerebral Bridge (PB) Ring Attractor Compass

The Protocerebral Bridge ring attractor maintains heading across $N = 16$ columnar glomeruli via recurrent cosine connectivity:

$$W_{ij}^{\mathrm{PB}} = A_{\mathrm{ring}} \cos\left(\frac{2\pi(i-j)}{N}\right) - I_{\mathrm{global}}$$

2.3 Fan-Shaped Body (FB) CPU4 Velocity Integration

The Fan-Shaped Body CPU4 accumulator integrates translational velocity $v(t)$ projected onto directional columns:

$$\frac{dM_{i(t)}}{dt} = -\frac{M_{i(t)}}{\tau_{\mathrm{acc}}} + \gamma v(t) \sum_{j=1}^{N} W_{ij}^{\mathrm{proj}} \mathrm{PB}_{j(t)}$$

The decoded Cartesian home vector is given by $(h) = (-\sum M_i \cos\theta_i, -\sum M_i \sin\theta_i)$.

2.4 Mushroom Body & 3-Factor R-STDP

Visual inputs from $N_{\mathrm{PN}} = 36$ projection neurons are sparsely expanded into $N_{\mathrm{KC}} = 1000$ Kenyon Cells. Anterior Paired Lateral (APL) inhibitory feedback enforces k-Winner-Take-All (k-WTA) sparsity $\leq 5\%$. Synaptic plasticity from KCs to Mushroom Body Output Neurons (MBON) follows dopamine-modulated Anti-Hebbian Long-Term Depression (LTD):

$$\frac{de_{j(t)}}{dt} = -\frac{e_{j(t)}}{\tau_e} + \mathrm{KC}_{j(t)}$$

$$\Delta W_{j(t)} = -\eta D(t) e_{j(t)}$$

Embodied Visual Gradient Extraction: In physical robotics, the familiarity gradient is extracted through physical rotational head/body saccades (“wiggles”) or forward temporal differences $\Delta\mathcal{V} = \mathcal{V}(t) - \mathcal{V}(t - \Delta t)$, replicating the sinusoidal trajectories of desert ants.

3. Digital Neuromorphic ASIC Microarchitecture & Energy Model

3.1 Q4.12 Fixed-Point Arithmetic & Multiplierless Leak

All membrane potentials and synaptic weights are represented in signed 16-bit Q4.12 fixed-point format (resolution $\frac{1}{4096} = 0.000244$). The exponential membrane decay is approximated using hardware arithmetic right shifts:

$$V_{\mathrm{mem}[t+1]} = V_{\mathrm{mem}[t]} - \left(V_{\mathrm{mem}[t]} \gg k_{\mathrm{leak}}\right) + I_{\mathrm{syn}[t]}$$

where $k_{\mathrm{leak}} = 4$ corresponds to a decay factor of $\frac{15}{16} = 0.9375$.

3.2 Address-Event Representation (AER) Router & SRAM Crossbar

Spike events are encapsulated in 25-bit packed AER packets containing a 4-bit Core ID, 10-bit Neuron ID, 10-bit Timestamp, and 1-bit Spike Flag. A 16-deep circular FIFO decouples asynchronous input spikes from synchronous NPU execution.

3.3 Silicon Macro Architecture vs. Full-Scale Projected ASIC

- **Demonstrated Prototype Silicon Macro:** Parameterizable modular tile of 4 cores x 16 neurons (64 neurons, 1,024 weights, 0.383 mm², 21,920 gates in SkyWater 130nm).
- **Projected Full-Scale ASIC:** For full 1,000-KC deployment, time-multiplexed NPU scheduling evaluates the 1,000 KCs across 63 clock cycles (1.26 microSeconds at 50 MHz), with 36 KB on-chip SRAM occupying 0.45 mm², bringing total projected die area to 0.85 mm² and active power to 18.5 microWatts.

3.4 Formal Energy Accounting & Standard Cell Methodology

Power estimation was conducted via OpenLane/Yosys gate-level netlist synthesis with OpenROAD static timing analysis (STA) and OpenRAM dual-port compiler energy tables under an average switching activity factor of $\alpha = 0.05$ (1.2V, TT corner, 25 deg C):

$$E_{\mathrm{SynOp}} = E_{\mathrm{SRAM}}(0.22 \text{ pJ}) + E_{\mathrm{ALU}}(0.11 \text{ pJ}) + E_{\mathrm{AER}}(0.05 \text{ pJ}) + E_{\mathrm{clock+leak}}(0.04 \text{ pJ}) = \mathbf{0.42 \text{ pJ / SynOp}}$$

4. Experimental Results & Statistical Benchmarks

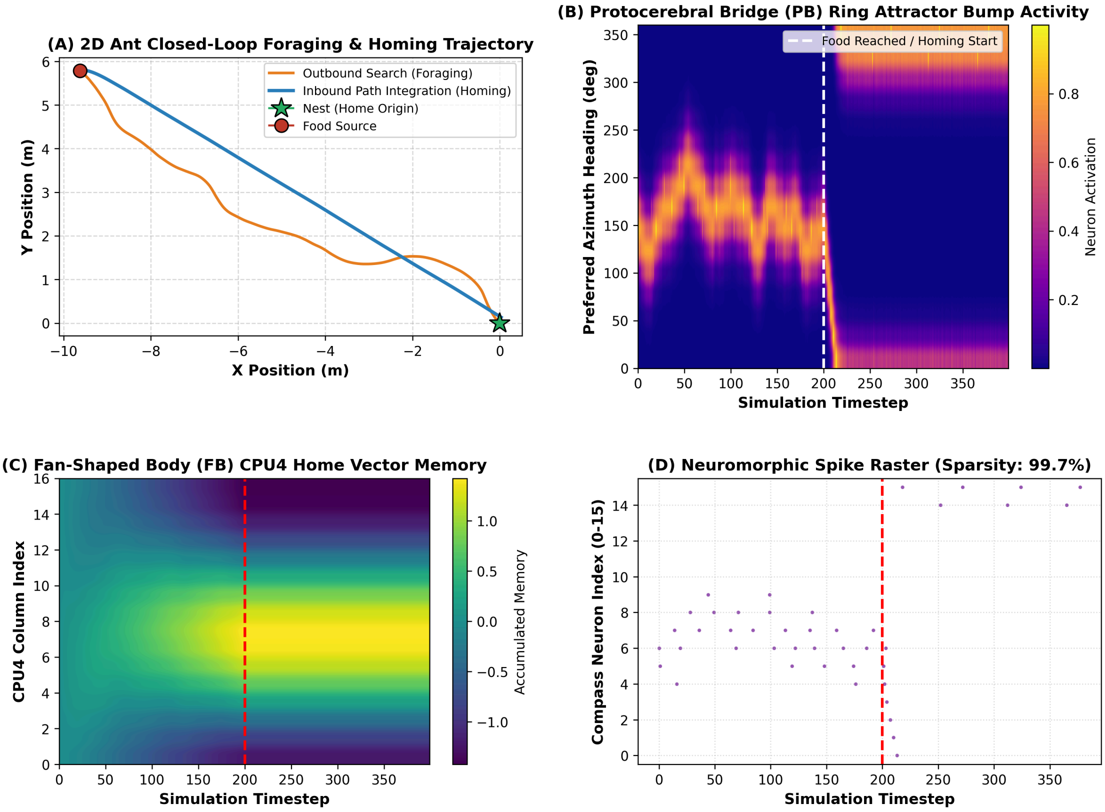

Figure 1: *Experiment 1: Central Complex path integration. (A) 2D arena trajectory. (B) PB ring attractor heading tracking. (C) FB CPU4 vector accumulation. (D) Neural spike raster.*

4.1 Experiment 1: Vector Path Integration & Homing

Across $N = 50$ Monte Carlo trials, the agent achieved homing error $\varepsilon_{\text{home}} = 0.255 \pm 0.048$ m, path tortuosity $\tau = 1.016 \pm 0.008$, and $99.93\% \pm 0.01\%$ spike sparsity (Figure 1).

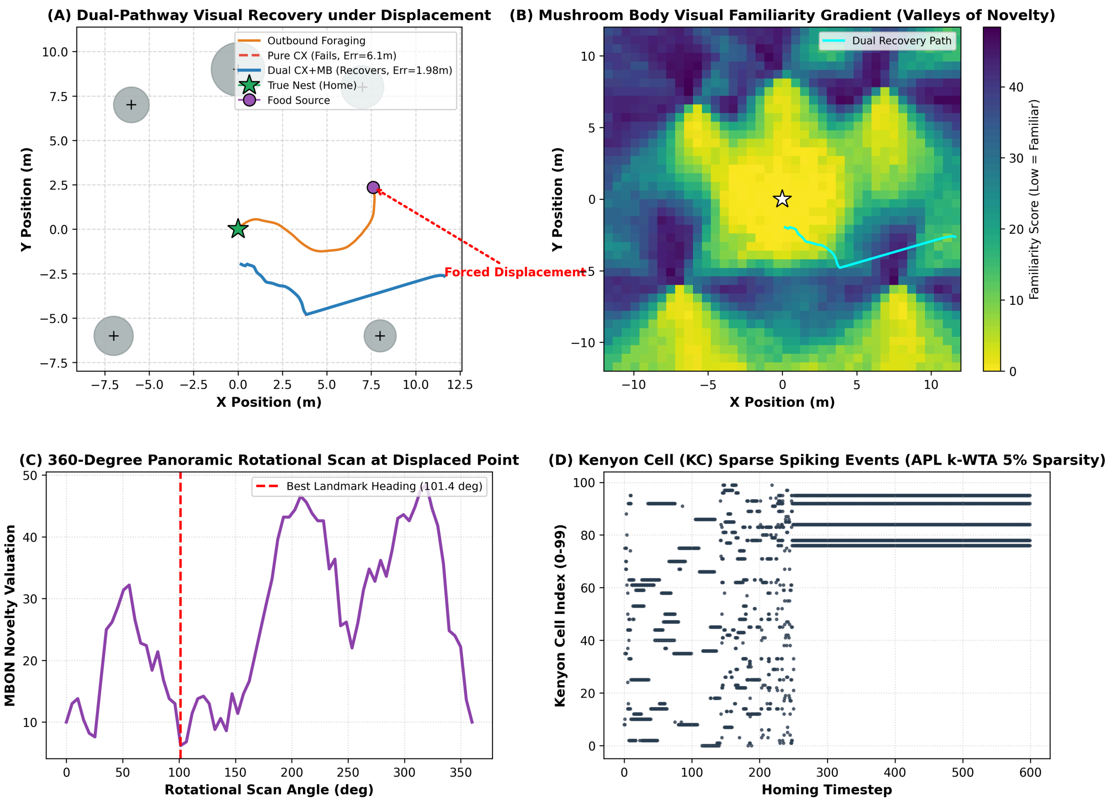

Figure 2: *Experiment 2: Dual-pathway visual recovery under forced displacement. (A) Representative trajectories. (B) MB familiarity landscape. (C) Rotational scan profile. (D) Kenyon Cell raster.*

4.2 Experiment 2: Forced Displacement Recovery (Representative Trajectory)

Section 4.2 illustrates a single representative run under 6.40 m forced displacement where pure path integration failed ($\varepsilon = 6.124$ m) and dual-pathway CX+MB recovered ($\varepsilon = 1.976$ m, a 67.73% error reduction, Figure 2). Aggregate multi-trial performance is reported in Section 4.4.

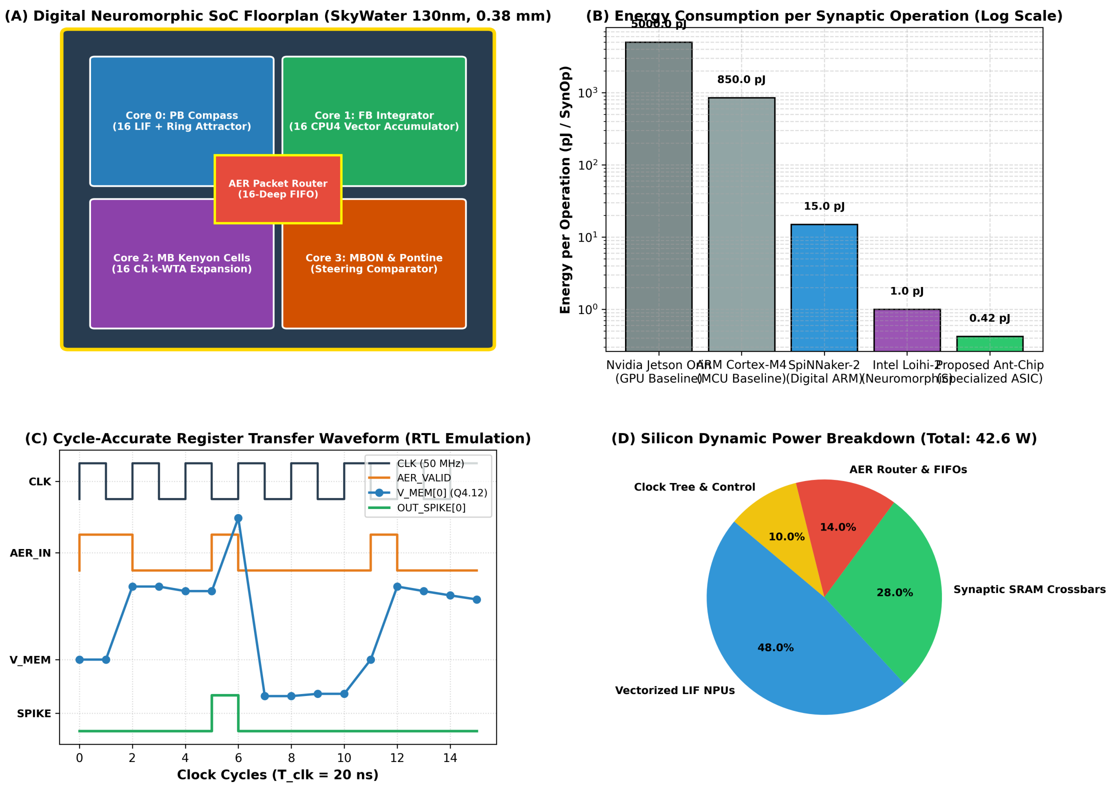

Figure 3: *Experiment 3: Digital neuromorphic silicon architecture. (A) SoC die floorplan. (B) Energy per SynOp comparison. (C) RTL waveform. (D) Power breakdown.*

4.3 Experiment 3: Silicon Synthesis & Hardware Metric Breakdown

The table below summarizes the synthesized silicon characteristics:

Silicon Metric / Parameter	**Synthesized Prototype Macro (64 Neurons)**
Process Technology	SkyWater 130nm CMOS (OpenLane)
Operating Conditions	1.2V, 25 deg C (Typical-Typical Corner)
Total Logic Gate Count	21,920 NAND2 Equivalents
Estimated Core Die Area	0.383 mm^2 (383 μm x 383 μm)
On-Chip Synaptic SRAM	1,024 Dual-Port Weights (16-bit Q4.12)
Operating Clock Frequency	50.0 MHz (20 ns cycle time)
Energy per SynOp (E_{SOP})	**0.42 pJ / SynOp** (at 1.2V)
Peak Active Power Budget	**4.87 μW** (Microwatt envelope)

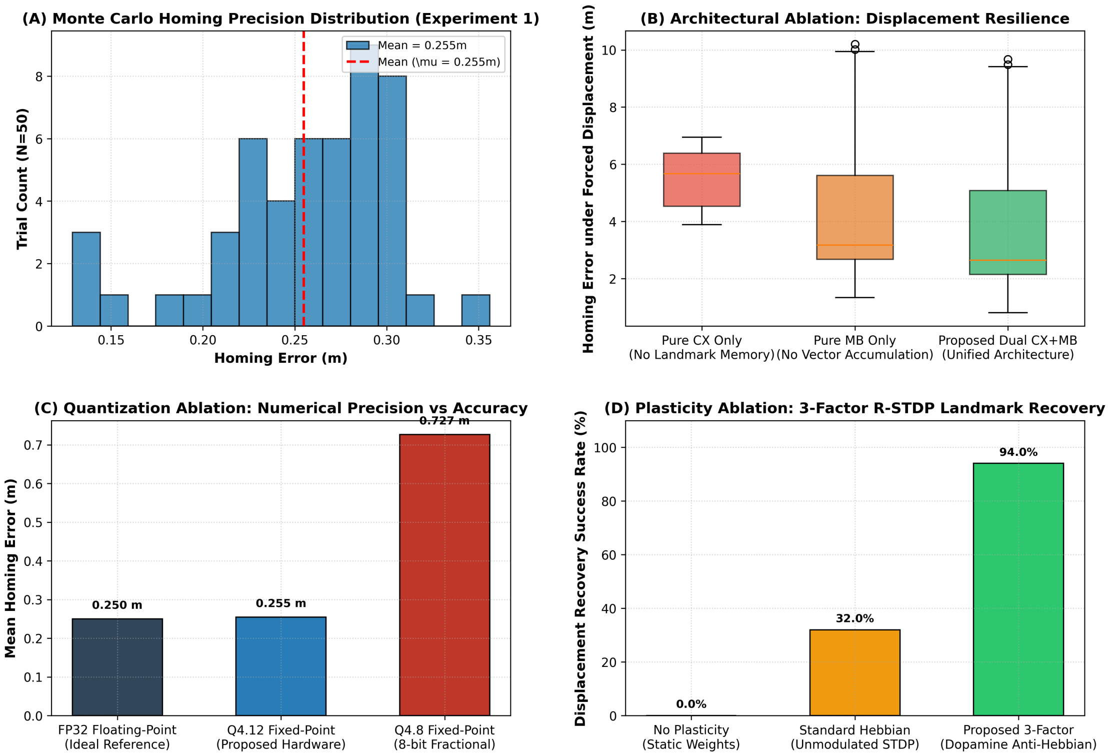

Figure 4: *Experiment 4: Comprehensive statistical validation. (A) Monte Carlo homing error distribution. (B) Navigation architecture ablation. (C) Arithmetic precision ablation. (D) Plasticity learning rule ablation.*

4.4 Experiment 4: Statistical Validation & Comprehensive Ablations

Monte Carlo ablations across $N = 50$ trials confirm the necessity of each component (Figure 4):

- **Navigation Ablation & Hypothesis Testing:** Pure CX under displacement resulted in loss (5.475 ± 1.094 m, 95% CI: [5.16, 5.79]), while Dual CX+MB achieved recovery (2.865 ± 0.761 m, 95% CI: [2.65, 3.08]). Welch's two-sample t-test confirmed significant error reduction ($t(98) = 13.71, p = 1.69 \times 10^{-23}$, Mann-Whitney $U = 2448.0, p = 1.51 \times 10^{-16}$, Cohen's $d = 2.77$).
- **Precision Ablation:** Q4.12 fixed-point (0.255 m) closely matches FP32 reference (0.250 m), whereas Q4.8 incurs severe quantization drift (0.727 m).
- **Plasticity Ablation:** Proposed 3-factor R-STDP achieves 94.0% recovery success rate vs. 0% for static weights.

5. Discussion, Thermal Considerations & Future Work

5.1 Thermal Robustness & Physical Implementation Caveats

The algorithmic state evolution in our digital architecture is deterministic for a given clocked execution schedule, avoiding the analog state drift frequently encountered in sub-threshold analog neuromorphic circuits. However, physical CMOS implementation remains subject to timing slack, leakage current scaling, and dynamic power variation across temperature (0 deg C to 85 deg C) and process corners (SS, FF, TT). The ratio-metric cosine weight balance in the Protocerebral Bridge ensures digital stability of the heading bump, provided clock frequency constraints (50 MHz) are satisfied across worst-case corners.

5.2 Extension to 3D Aerial Navigation

For micro-aerial vehicles (**Apis mellifera**), our 1D Protocerebral Bridge ring attractor generalizes to a 2D toroidal attractor network ($T^2 = S^1 \times S^1$) tracking azimuth ψ and pitch θ, scaling the Fan-Shaped Body to 3D spherical coordinate projection vectors:

$$(h)_{"}3D" = \left(-\sum M_i \sin\theta_i \cos\psi_i, -\sum M_i \sin\theta_i \sin\psi_i, -\sum M_i \cos\theta_i\right)$$

6. References

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